

For Loop Diagnostic test

Name (caps) _____ Student ID (9 digits)

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Lab Time _____ Desk Number: Desk-_____ (eg Desk-G02-060)

How to work out what a code fragment does: draw a table

The simplest and most reliable way to work out what a code fragment does (and to understand) is to draw a table. This is called tracing or dry running code, Let's work out what this fragment does:

```
for (int i = 1; i <= 2; i++) // L1
{
    System.out.println(i);    // L2
}
System.out.println("DONE");    // L3
```

Executing Line L1

First the counter variable i is initialised to 1. Then the loop is evaluated to see if the loop terminates. The test is true so we enter the loop body.

Executing Line L2

The loop body prints i, which is 1.

Executing Line L1

When the body is executed we return back to the for line L1 and execute the statement updating i (i++). It adds 1 to i, so i becomes 2 and the test is checked again. It is still true, so we enter the body and execute line L2 again.

Executing Line L2

The loop body prints i, which is now 2.

LINE	i	i<=2?	Printed
L1	1	1<=2? true	
L2			1
L1	2	2<=2? true	
L2			2

Executing Line L1

We return back to line L1 and update i again, so it becomes 3 and the test is checked again. It is now false so the loop terminates and jumps to the next line, L3

Executing Line L3

This just prints DONE

Overall what is printed is:

1
2
DONE

LINE	i	i<=2?	Printed
L1	1	1<=2? true	

LINE	i	i<=2?	Printed
L1	1	1<=2? true	
L2			1

LINE	i	i<=2?	Printed
L1	1	1<=2? true	
L2			1
L1	2	2<=2? true	

LINE	i	i<=2?	Printed
L1	1	1<=2? true	
L2			1
L1	2	2<=2? true	
L2			2
L1	3	3<=2? false	

LINE	i	i<=2?	Printed
L1	1	1<=2? true	
L2			1
L1	2	2<=2? true	
L2			2
L1	3	3<=2? false	
L3			DONE

You must show all your working in the following. No correct table (exactly the same as the solution)... no marks.

1. What is printed when the following code fragment has executed?

```
int a = 3; // L1
for (int i = 5; i <= 7; i++) // L2
{
    a = a + 7; // L3
    System.out.println(a); // L4
}
System.out.println(a+1); // L5
```

Printed:

LINE	a	i	i<=7 ?	Printed

2. What is printed when the following code fragment has executed?

```
int a = 3; // L1
for (int i = 5; i < 7; i++) // L2
{
    a = a + 7; // L3
    System.out.print(a); // L4
}
System.out.print(a+1); // L5
```

Printed:

LINE	a	i	i<7 ?	Printed

3. What is printed when the following code fragment has executed?

```
int a = 4; // L1
for (int i = 0; i <= 1; i++) // L2
{
    System.out.println(a); // L3
    a = a + 5; // L4
}
System.out.println(a); // L5
```

Printed:

LINE	a	i	i<=1 ?	Printed

4. What is printed when the following code fragment has executed?

```
int a = 4; // L1
for (int i = 0; i <= 2; i++) // L2
{
    a = a + 5; // L3
}
System.out.println(a); // L4
```

Printed:

LINE	a	i	i<=2 ?	Printed

5. What is printed when the following code fragment has executed?

```
int a = 4; // L1
for (int j = 0; j < 0; j++) // L2
{
    a = a + 5; // L3
}
System.out.println(a); // L4
```

Printed:

LINE	a	j	j<0 ?	Printed

6. What is printed when the following code fragment has executed?

```
int a = 4; // L1
for (int j = 0; j <=0; j++) // L2
{
    a = a + 5; // L3
    System.out.println(a); // L4
}
```

Printed:

LINE	a	j	j<=0 ?	Printed

7. What is printed when the following code fragment has executed?

```
int b = 3; // L1
for (int i = 7; i >= 5; i--) // L2
{
    b = b + 7; // L3
    System.out.println(b); // L4
}
```

Printed:

LINE	b	i	i>=5 ?	Printed

8. What is printed when the following code fragment has executed?

```
int b = 10; // L1
for (int i = 1; i < 5; i=i+2) // L2
{
    b = b + 10; // L3
    System.out.println(b); // L4
}
```

Printed:

LINE	b	i	i<5 ?	Printed